

VENKATRAMAN R

Data Scientist · Machine Learning Engineer · AI Engineer

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SUMMARY

Entry-level Data Scientist fluent across the full machine learning lifecycle — statistics, algorithms, applied mathematics (linear algebra, calculus), modelling, explainability, data analytics, data visualization, and deployment — demonstrated through 8 production-grade projects shipped end to end. Strong in Python, SQL, machine learning (XGBoost, Random Forest, Scikit-Learn), deep learning (TensorFlow, Keras, LSTM), and generative AI (LangChain, FAISS, RAG); deployed models reach AUC-ROC 0.97 with audit-ready SHAP explainability. Analytical researcher with strong communication skills and problem-solving instincts who offsets a non-traditional academic path with a deep, shipped portfolio. Seeking Data Scientist, Machine Learning Engineer, or AI Engineer roles.

TECHNICAL SKILLS

Programming	Python, SQL
Math & Stats	Statistics, Linear Algebra, Calculus, Applied Mathematics, Probability, Algorithms, Hypothesis Testing, Analytical Methods, Mathematical Software, Business Rules
Machine Learning	Scikit-Learn, XGBoost, Random Forest, SMOTE, SHAP, Optuna, Cross-Validation, Model Evaluation, Feature Engineering
Deep Learning	TensorFlow, Keras, LSTM, Neural Networks, RNN, CNN (foundational)
GenAI / LLMs	LangChain (LCEL), FAISS, RAG, OpenAI API, Groq Llama-3.3-70B, HuggingFace, Prompt Engineering, NLP
Analytics & Viz	Pandas, NumPy, Power BI (DAX), Plotly, Matplotlib, Seaborn, Data Visualization, Streamlit, SQLite, Parquet, Power Query
Tools	Git, GitHub, Jupyter, VS Code, Google Colab, Advanced Excel

PROFESSIONAL EXPERIENCE

Data Science Trainee Jan 2026 – Jun 2026
Boston Institute of Analytics — Professional Certification Program

- Researched and built **FraudGuard** (institute flagship capstone): trained a machine learning classifier (XGBoost) with Optuna hyperparameter tuning (100 trials) and SMOTE balancing on imbalanced mobile-transaction data, reaching AUC-ROC 0.97, F1 0.92, precision 0.93, recall 0.91; encoded fraud-cost asymmetry as business rules via F2-score threshold optimisation and added SHAP per-transaction explanations for audit compliance
- Designed **BankShield** capstone: an end-to-end machine learning pipeline (Random Forest; EDA → feature engineering → SMOTE → threshold tuning → deployment) reaching AUC-ROC 0.97, precision 0.94, recall 0.89; surfaced transaction velocity as more predictive than transaction amount; deployed a Streamlit app with single and batch CSV scoring and presented findings to non-technical reviewers

PROJECTS

AskMyData — GenAI Dataset Q&A Application Live Demo · GitHub
Python · LangChain (LCEL) · FAISS · Groq Llama-3.3-70B · OpenAI · Streamlit

- Built a LangChain LCEL agent with a natural language (NLP) interface that auto-selects pandas or SQL to answer plain-English questions across 4 source formats (CSV, Excel, JSON, SQLite), with FAISS retrieving per-column schema context for accuracy
- Engineered a secure execution sandbox (whitelisted builtins, auto-retry), multi-turn memory, auto-generated Plotly visualizations per query, and a live Streamlit Cloud deployment

SalesPulse — Power BI Sales Intelligence Dashboard GitHub
Power BI · DAX · Power Query · Star Schema · Data Visualization

- Authored 44 custom DAX measures (YoY, MTD/QTD/YTD, customer LTV, profit margin) over a star schema on \$12.63M of Global Superstore data, with executive, regional, product, and customer-segment drill-throughs

SalesPulse Forecaster — LSTM 12-Month Sales Forecasting GitHub
TensorFlow · Keras · LSTM · SQLite · Streamlit · Statistics

- Built a 2-layer deep learning LSTM network (128 → 64 units) generating recursive 12-month forecasts with confidence bands; backed by a SQLite pipeline and a Streamlit upload-to-export interface

DataHunt — Automated ETL Pipeline GitHub
Python · Pandas · Parquet · openpyxl · Algorithms · Logging

- Designed a modular pipeline (Extractor → Cleaner → Storage → Reporter) outputting Parquet and Excel simultaneously, with fully configurable strategies and independently swappable modules

Vehicle Maintenance Dashboard — Fleet Analytics Advanced Excel
Built a fleet analytics dashboard over 50,000-vehicle maintenance records with KPI tracking and conditional-formatting-driven alerts

EDUCATION

Professional Certification — Data Science with Artificial Intelligence Jun 2026
Boston Institute of Analytics

Coursework: Statistics, Linear Algebra, Calculus, Probability, Algorithms, Machine Learning, Deep Learning, NLP, Generative AI & RAG, Power BI, SQL, EDA, Data Visualization. Capstones: FraudGuard (institute flagship) & BankShield.

Bachelor of Engineering (B.E.) — Automobile Engineering 2021 – 2024
Vels Institute of Science, Technology & Advanced Studies (VISTAS) · CGPA 7.25